EE121 Fall 2025 | Miki Lustig
Digital Communications Systems

Assignment 7

Due Sunday November 2nd, 2025, Self Grading Due Sunday November 9th, 2025

1. You are given a channel bandwidth of 3 kHz. The requirement is to transmit data over the channel at a rate of 4.5 kbits/s using binary PAM.
   (a) What is the maximum roll-off factor in the raised-cosine pulse spectrum that can accommodate this data transmission?
   (b) What is the corresponding excess bandwidth?
2. The binary sequence 11100101 is applied to a QPSK modulator. The bit duration is 1 µs. The carrier frequency is 6 MHz.
   (a) Calculate the transmission bandwidth of the QPSK signal.
   (b) Plot the waveform of the QPSK signal.
3. Madhow Problem 4.15
4. **Self-grade Homework 6.** Fill out the self-grade rubric for Homework 6. Attach to Homework 7 submission.